

Showkot Hosen

XAI in Cybersecurity Researcher

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LinkedIn Portfolio GitHub Latifpur, Chittagong, Bangladesh

About

A strong research-oriented interest in cybersecurity has been developed, with a particular focus on explainable artificial intelligence (XAI) for secure and trustworthy cyber defense systems. Solid foundations in computer networking, ethical hacking, and system security have been established through academic training and hands-on projects. Current research interests center on the design and implementation of machine learning-based intrusion detection and prevention systems with enhanced interpretability to support transparent and reliable security decisions. A continuous commitment is maintained to advancing knowledge in emerging cyber threats, adversarial techniques, and explainable security analytics in preparation for doctoral research.

Expertise, Skills & References

Area of Expertise	Technical Skills	References
Cybersecurity	Languages: Python, Bash, C++	Dr. Md. Azad Hossain
Machine Learning	Security Tools: Wireshark, Nmap, Nikto, Ettercap	Prof., Dept. of ETE, CUET azad@cuet.ac.bd +8801783756981
Explainable AI (XAI)	AI Tools: Claude, ChatGPT, Gemini, Perplexity	Dr. Md. Nursadul Mamun
Research & Experimentation	Version Control and IDE: Github, VS Code, Arduino IDE	Prof., Dept. of ETE, CUET nursad.mamun@cuet.ac.bd +8801813977506
Prompt Engineering	Hardware: Arduino, ESP32	

Education

MSc in Electronics and Telecommunication Engineering Chittagong University of Engineering and Technology (CUET)	August 2026 – Present Chattogram, Bangladesh
- Major: Artificial Intelligence and Cybersecurity - Research focus: XAI in Cybersecurity, Intrusion Detection Systems (IDS), Machine Learning, and LLM Security	
BSc in Electronics and Telecommunication Engineering Chittagong University of Engineering and Technology (CUET)	March 2022 – June 2026 Chattogram, Bangladesh
- CGPA: 3.63 / 4.00 - Research focus: XAI in Cybersecurity, Intrusion Detection Systems (IDS), and Machine Learning	
Higher Secondary Certificate (HSC) Government City College GPA: 5.00 / 5.00	2018 – 2020 Chittagong, Bangladesh
Secondary School Certificate (SSC) Latifpur Alhaj Abdul Jalil High School GPA: 5.00 / 5.00	2016 – 2018 Chittagong, Bangladesh

Publications

A Duplicate-Audited, SHAP-Explainable Decision Tree for Lightweight Intrusion Detection in LEACH-Based WSNs | Q2
Submitted to the Journal of Cyber Security Technology, Taylor & Francis
Artifacts: Zenodo DOI 10.5281/zenodo.22066348

PromptLight: A Lightweight, Adversarially Robust, and Explainable Four-Layer Prompt-Injection Detector | Q1

Undergraduate Thesis work Submitted to Cybersecurity and Applications

[View Visual Abstract](#)

Interpretable SQL Injection Detection: Lightweight Decision Trees with SHAP-Enhanced Deployment | QPAIN 2026

2026 IEEE 2nd International Conference on QPAIN

[View Publication](#)

XAI-Powered Vision: Making CNN Malware Classifiers Transparent for Cybersecurity | QPAIN 2026

2026 IEEE 2nd International Conference on QPAIN

[View Publication](#)

Aircraft Detection in Satellite Images Using CNN and Sliding Window Method | ECCE 2025

2025 International Conference on ECCE, CUET, Bangladesh

[View Publication](#)

Academic Service

Invited Reviewer

2026

Cyber Security and Applications (Elsevier)

- Invited by the editorial board to serve as a peer reviewer for manuscripts submitted to the journal.

Reviewer Panel Member & Invited Reviewer

2026

HKIE Transactions – The International Journal of The Hong Kong Institution of Engineers (HKIE)

- Added to the journal's reviewer panel and invited to review manuscripts within my area of expertise.

Experiences

Adjunct Lecturer

2026 – Present

International Islamic University Chittagong (IIUC)

Chattogram, Bangladesh

- Teach undergraduate courses in the Department of Electronic and Telecommunication Engineering (ETE).
- Conduct lectures, practical lab sessions, and mentor undergraduate students in electronics and telecommunication engineering topics.

Cybersecurity Research Student

2025 – Present

CUET

Chattogram, Bangladesh

- Conducted hands-on research on IDS/IPS and ML-based cyber defense systems.
- Participated in Capture The Flag (CTF) competitions and OSINT investigations.

Cyberhunt – Cybersecurity Mentorship Lead

2024 – Present

CUET

Chattogram, Bangladesh

- Mentor and guide departmental junior students in the fields of Cybersecurity and Machine Learning.
- Inspire students to pursue careers in ML-based cybersecurity research and development programs.

Certifications & Credentials

S.	Certification	Vendor
1	DevSecOps ↗	Phoenix
2	Advent of Cyber ↗	THM
3	Pre-Security ↗	THM
4	Network Analysis ↗	Blue Team
5	Certified in Cyber ↗	(ISC)2
6	CC Specialization ↗	(ISC)2
7	Ethical Hacker ↗	Cisco
8	Network Devices ↗	Cisco
9	OS Basics ↗	Cisco
10	Intro to Cyber ↗	Cisco
11	Intro to IoT ↗	Cisco

S.	Certification	Vendor
12	Network Basics ↗	Cisco
13	Hardware Basics ↗	Cisco
14	Critical Infra ↗	OPSWAT
15	ML & Emerging Tech ↗	Coursera
16	AI for Cyber ↗	Coursera
17	PenTesting & Crypto ↗	Coursera
18	Python Basics ↗	BCC
19	Cyber Job Sim ↗	Deloitte
20	Cyber Job Sim ↗	Mastercard
21	Shields Up Sim ↗	Forage
22	Cyber Professional ↗	Diligite

Cybersecurity & ML Projects

1.

SEC-IoT – A Real-Time Secure Industrial IoT Monitoring & Analytics Platform
Python, HTML, CSS, C++, JAVA, ChatGPT, Gemini AI
2.

MLWAF – A Real-Time Machine Learning Powered Web Application Firewall
Python, Scikit-learn, Network Traffic Analysis. Designed supervised ML models for detecting malicious network traffic patterns.
3.

Sentrix: Smart Industrial Safety
ESP32, ESP8266, YOLO, Computer Vision. Industrial IoT framework for real-time safety monitoring, facial recognition, and automated hazard controls.
4.

IoT Military Monitoring System
ESP-12, Arduino, RFID, Bio-Sensors. Multi-sensor system with access control, soldier health tracking, and environmental safety detection.
5.

Text Steganography Detection Using Deep Learning
Built and trained in the Kaggle ML environment. Performed feature extraction and deep neural network optimization.
6.

Ethical Hacking Capstone CTF
SQL Injection, SMB Exploitation, PCAP Analysis. Solved real-world attack scenarios including web exploitation.
7.

Chrome Extension Malware Analysis & Removal
JavaScript, Security Analysis. Identified malicious extensions using permission and behavior analysis.
8.

Quranic Shots: Islamic Accessories E-Commerce Site
HTML, CSS, PHP, JS, XAMPP. Developed a full-stack e-commerce platform with secure web environment testing.
9.

Professional Portfolio Site
HTML, CSS. Designed and deployed a personal portfolio using W3Spaces to showcase technical projects.
10.

Infrared Communication System (Electronics Project)
A full hardware-based communication system utilizing infrared signals for wireless data transmission.

Co-Curricular Activities

- Phoenix Summit 2026 Fellowship

Dhaka, Bangladesh

DevSecOps Fellow

- Selected for an intensive 3-day DevSecOps fellowship focused on modern security integration.
 - Gained hands-on training in securing CI/CD pipelines and automated security deployment strategies.
- Organizer | Televerse 1.0 (CTF Event)

CUET, Bangladesh

Technical Lead & Event Coordinator

- Managed a national-level cybersecurity CTF competition at CUET.
 - Led the technical team in designing defensive security challenges and coordinated logistics for 50+ participants.
- Industrial Attachment | DigidenIT

Chattogram, Bangladesh

Web Development & IT Infrastructure Trainee

- Gained hands-on experience in networking fundamentals, system administration, and web security.
 - Assisted in real-world IT support environments and infrastructure maintenance tasks.
- Competitive Achievements

Inter-University/National

Case Solving,CTF and ML

- **National Top 16:** Nscania AI Hackathon (National AI Competition).
 - **National Rank 37th:** BUET DL Sprint 4.0 (Bengali Speech Recognition ML Competition).
 - **Semifinalist:** Face The Case 2.0 (Business Case Competition).
 - **Participant:** BCS CTF and UAP CTF (Cybersecurity Capture The Flag Competitions).